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WPC POWER DRIVER BOARD – COST–OPTIMISED MODERN VERSION – functional replacement for Williams A–12697–3 (same outline, connectors, fuses).
DPAK MOSFET solenoid/lamp drivers, TPS54360B buck regulators for +5V/+12V, 4–quadrant insulated triacs, SMD passives, 5x20 mm fuses.
Sheets: Power_Supply, CPU_Interface, Sol_HighPower (1–8), Sol_LowPower (9–16), Sol_Flashers (17–24), Sol_GeneralPurpose (25–28),
Lamp_Columns, Lamp_Rows, GI_Triacs. Every part carries Manufacturer / MPN / DigiKey link / Price / Price100 fields (see BOM). See README.md.

Power_Supply

File: power_supply.kicad_sch

CPU_Interface

File: cpu_interface.kicad_sch

Sol_HighPower

File: sol_highpower.kicad_sch

Sol_LowPower

File: sol_lowpower.kicad_sch

Sol_Flashers

File: sol_flashers.kicad_sch

Sol_GeneralPurpose

File: sol_gp.kicad_sch

Lamp_Columns

File: lamp_columns.kicad_sch

Lamp_Rows

File: lamp_rows.kicad_sch

GI_Triacs

File: gi_triacs.kicad_sch

Modern re–implementation of the Williams A–12697–3 (same form factor)

Sheet: /
File: wpc_power_driver_cost.kicad_sch

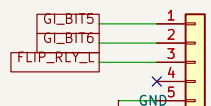
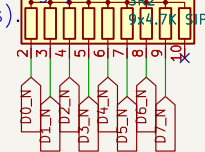
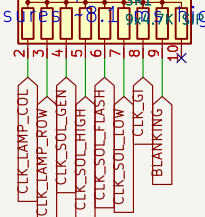
Title: WPC Power Driver Board – cost–optimised modern re–implementation

Size: A3Date: 2026–09–04Rev: A

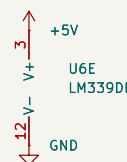
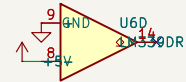
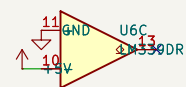
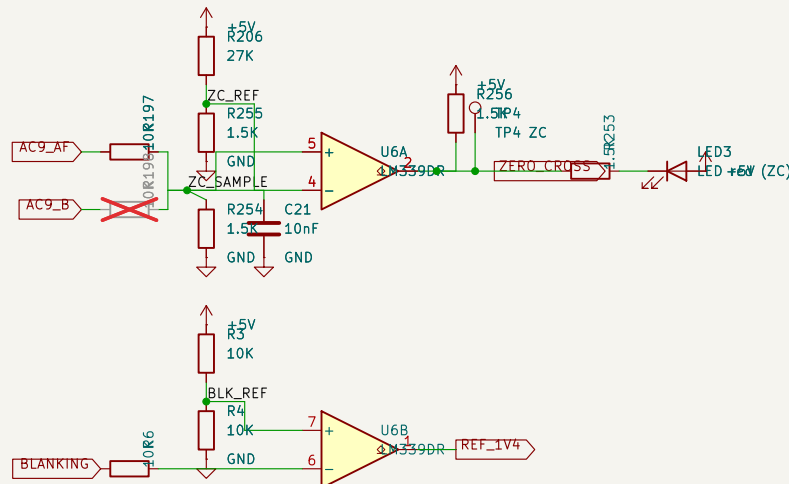
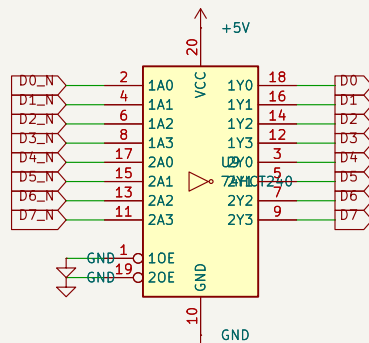
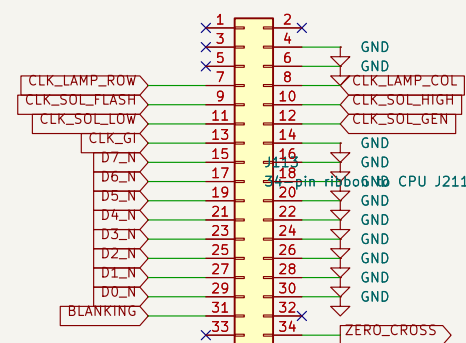
KiCad E.D.A. 10.0.6Id: 1/10

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12 CPU interface, pin-out from the WPC schematic manual 16-9834-2, J113: 7 /LMP ROW, 8 /LMP COL, 9 /SOL2 2 (sol 17-24), 10 /SOL 4 (sol 1-8), 11 /SOL 3 (sol 9-16), 12 /SOL 1 (sol 25-28), 13 /TRIAC, 15, 29 end = /DTL+/D0, 31 BLANKING (HIGH = every latch bit *tri-stated*, open-collector from the CPU, pulled up here), 34 ZERO CROSS to the CPU: 1/3/5/32/33 (Hilper-opto sense of the -1 relay board) and 2 are N/C on the A-12697-3; other even pins = ground. The CPU board drives the data through a 74LS240, so the ribbon data is ACTIVE LOW: U9 (74HCT240) inverts it and every latch then takes the true bus (register bit = 1 $\rightarrow$ output ON, as PinMAME / FreeWPC write it). BLANKING tri-states every 74HCT574 (gate pull-downs $\rightarrow$ all OFF) and pulls the lamp-row 1.4 V reference low. Zero-cross: the F113-fused leg of the 9 VAC winding is sampled through R197 /K0 / R254 1.5K / C21 (R498, DNP) into a +input of U6A against the 0.26 V reference on $\rightarrow$ ZERO CROSS is HIGH while that leg is positive: a 60 Hz ~~280 mV~~ ^{280 mV} wave (-7.7 mV half-cycle) with one edge per mains zero crossing $\rightarrow$ the waveform of the A-12697 (U6A pin 7, D38 sample: a real TP4 measure: 280 mV high pulses).

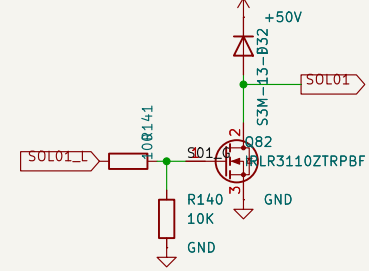


J111 G.I. latch spare bits T5-T7 (GPIO as on the original board, e.g. shot-clock drive; N/C on STTG

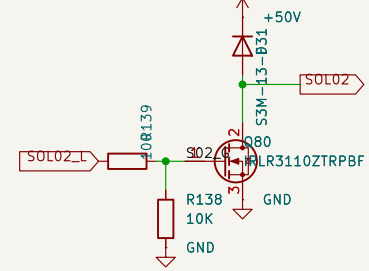


High power solenoid drivers (sol. 1–8): IRLR3110Z DPAK (100 V, 14 mOhm, 16 mOhm at 4.5 V) driven directly by latch U5 (74HCT574, output HIGH = ON, outputs tri–stated by BLANKING → gate pull–downs hold every driver OFF). SMC 3 A tie–back diodes return to +50V (the “+50V” rail of a WPC machine is ~70 V DC; 51 VAC winding, 72–78 V peak). U5 is clocked by /SOL 4 (J113–10). Output connectors J130 (playfield, key pin 3), J132 (backbox, sol 1–4, key 4) and J131 (cabinet, sol 5–8, key 2) as on the A–12697. Supply +50V via F105 on J107–3.

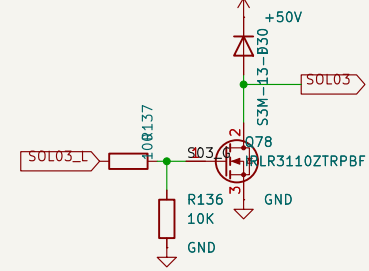
Sol 1 (high power)



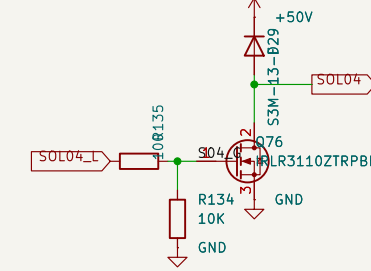
Sol 2 (high power)



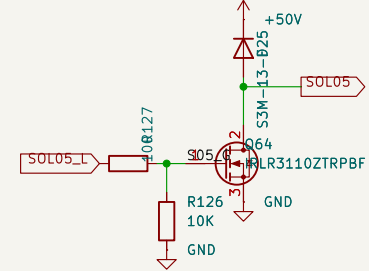
Sol 3 (high power)



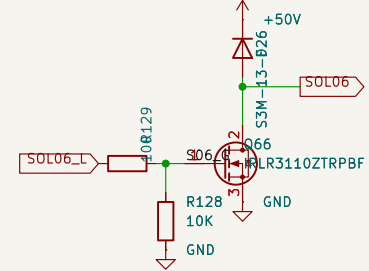
Sol 4 (high power)



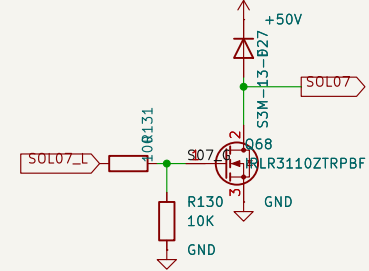
Sol 5 (high power)



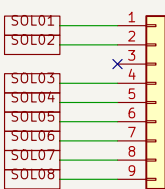
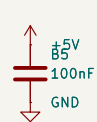
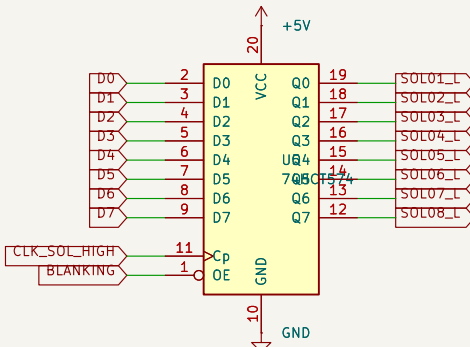
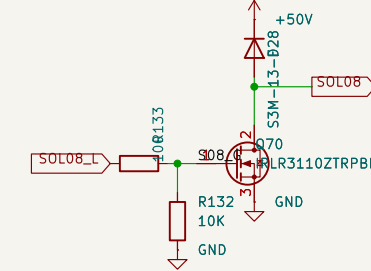
Sol 6 (high power)



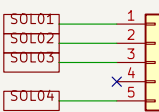
Sol 7 (high power)



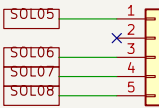
Sol 8 (high power)



J130 Sol 1–8 drive (playfield)



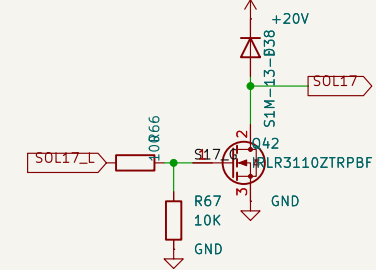
J132 Sol 1–4 drive (backbox, parallel to J130; N/C on STING)



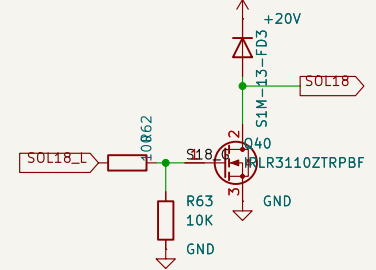
J131 Sol 5–8 drive (cabinet, parallel to J130; N/C on STING)

Flashlamp drivers (sol. 17-24): IRLR3110Z DPAK, latch U3 output HIGH = lamp ON, clocked by /SOL 2 (J113-9). Sol 17-20 keep an on-board S1M tie-back to +20V (the A-12697 has none there); the sol 21-24 tie-back cathodes are brought out to J126-10..13 exactly like the A-12697 D9-D12, so the harness chooses the clamp rail. On STING sol. 17/18 drive the 12 V gun motors (motor high side on the +12V power rail of J118) through this group. Outputs on J126 (playfield, key pin 9) and J125 (backbox, key pin 4). Flasher supply +20V on J107-5/6 / J106-5.

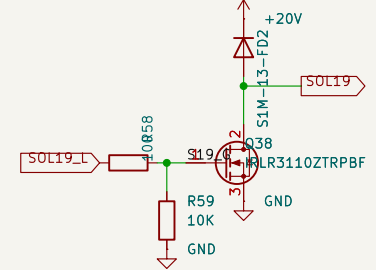
Sol 17 (flasher)



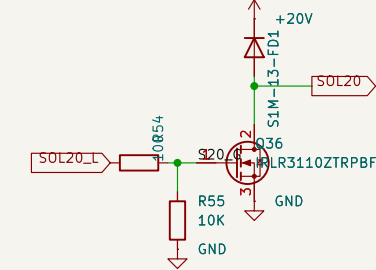
Sol 18 (flasher)



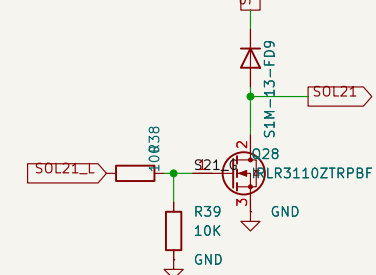
Sol 19 (flasher)



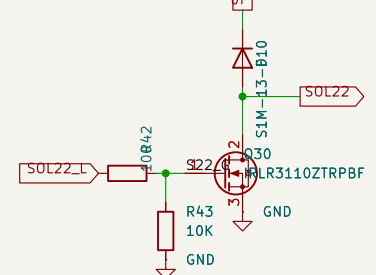
Sol 20 (flasher)



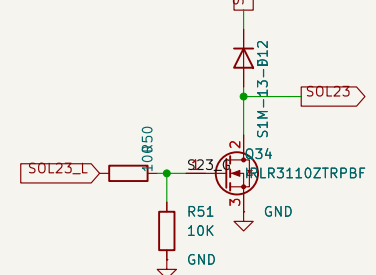
Sol 21 (flasher)



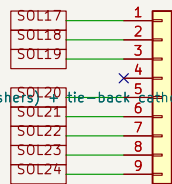
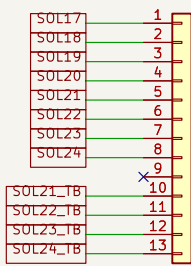
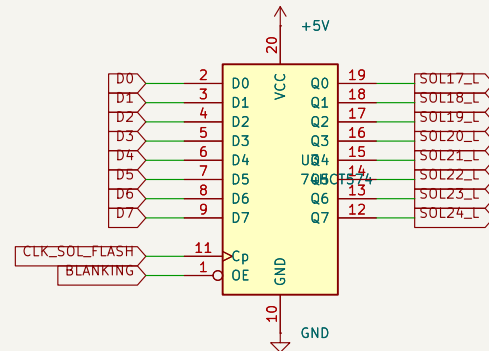
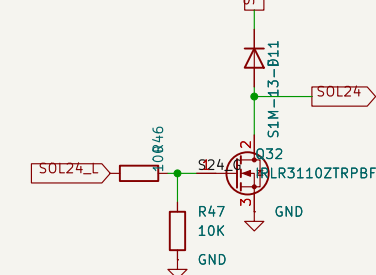
Sol 22 (flasher)



Sol 23 (flasher)

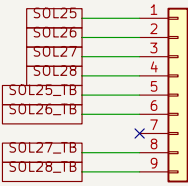
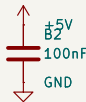
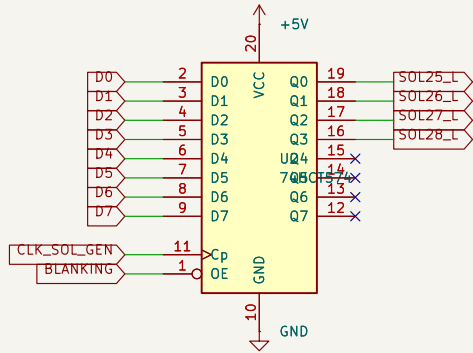
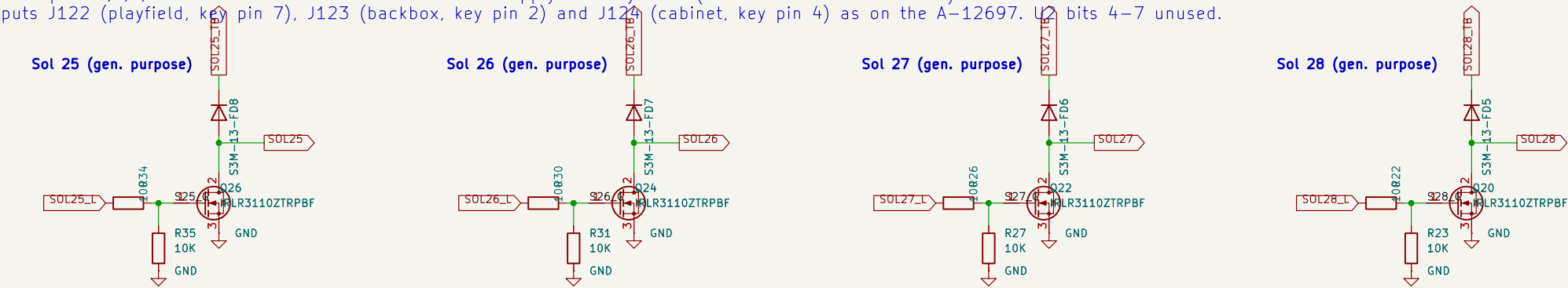


Sol 24 (flasher)

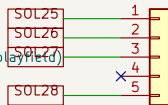


J126 Sol 17-24 drive (playfield flasher) J125 Sol 17-24 drive (backbox flasher)

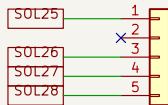
General purpose drivers (sol. 25–28): IRLR3110Z DPAK, latch U2 output HIGH = ON, clocked by /SOL 1 (J113–12). S3M tie-back diode cathodes are brought out to J122 pins 5,6,8,9 so the harness can tie them to the supply actually used (+50V coil or +20V flasher). Outputs J122 (playfield, key pin 7), J123 (backbox, key pin 2) and J124 (cabinet, key pin 4) as on the A–12697. U2 bits 4–7 unused.



J122
J122 Sol 25–28 drive + tie-back (playfield)



J124
J124 Sol 25–28 drive (backbox)



J123
J123 Sol 25–28 drive (backbox, parallel to J122/J124; N/C on STNG)

Modern re-implementation of the Williams A–12697–3 (same form factor)

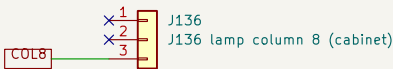
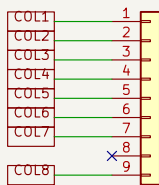
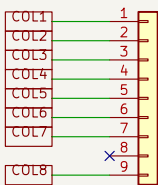
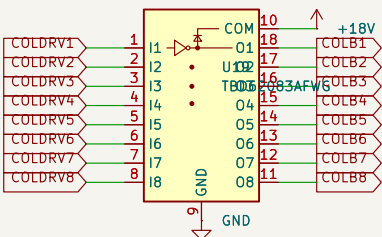
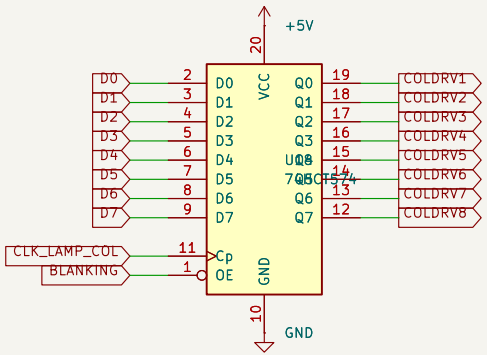
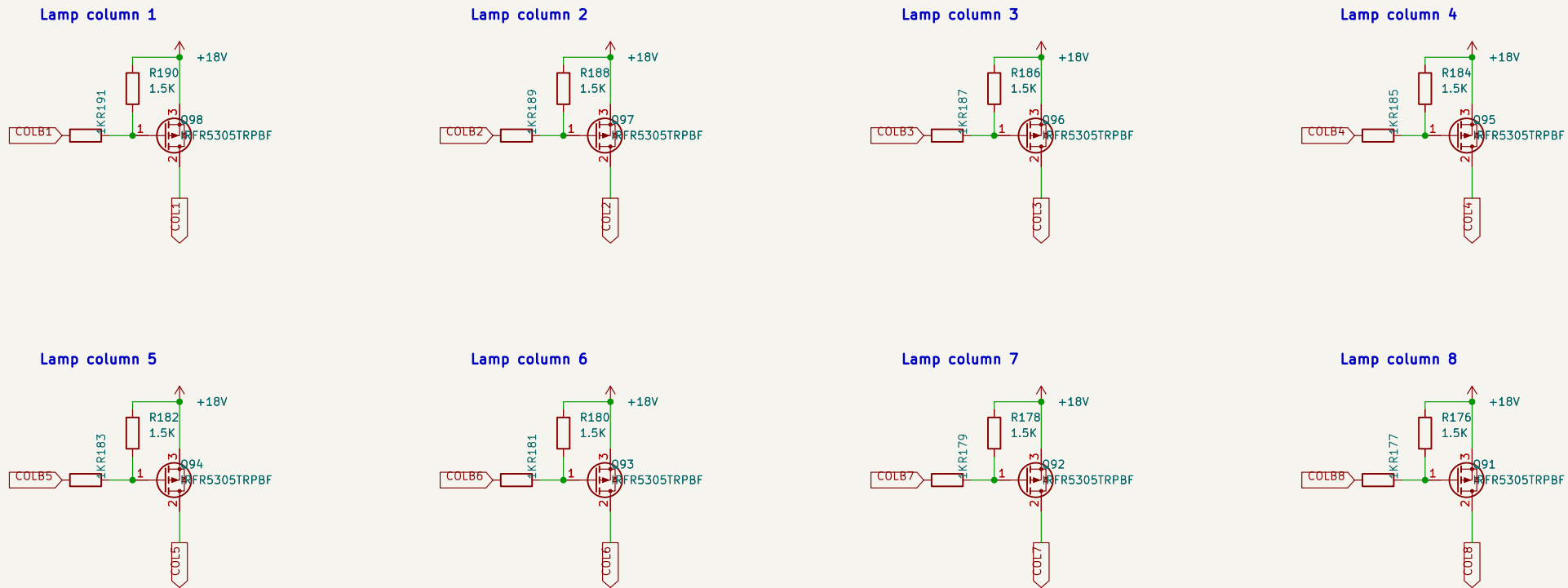
Sheet: /Sol_GeneralPurpose/
File: sol_gp.kicad_sch

Title: Solenoids 25–28 (general purpose, IRLR3110Z)

Size: A2
Date: 2026–09–04

Rev: A
Id: 7/10

Lamp matrix column drivers. Column strobe register 0x3FE5: bit(k-1)=1 selects column k, clocked by /LMP COL (J113-8). The TBD62083A inputs have an internal pull-down (Toshiba usage note), so a tri-stated U18 (BLANKING) leaves every column OFF – the OEM 74LS374/ULN2803 pair behaves the same way. U18 latches the data bus; the TBD62083A DMOS sink pulls the P-MOSFET gate down through 1 k against the 1.5 k gate-source resistor (Vgs = -0.6 x V+18 = -8..-12 V) -> column high (+18V) with -65 mOhm (IRFR5305 DPAK; TIP107 dropped -1.2 V). Column k = Q(99-k): col1=Q98 ... col8=Q91. Outputs J137/J138 pins 1-7,9 (key pin 8).



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /Lamp_Columns/
File: lamp_columns.kicad_sch

Title: Lamp matrix columns (IRFR5305 high side, +18V)

Size: A2
Date: 2026-09-04
KiCad E.D.A. 10.0.6

Rev: A
Id: 8/10

[illegible]

General illumination: each 6.3VAC string is fused (F106–F110) on the hot side and switched by a triac in the RETURN path; MT1 of the five triacs form the G_LRET bus that goes back to the transformer on J115–7/8/10/11/12 and is referenced to logic ground by W1 (OEM: J115–1 GND REF), so the 5 x 5 A return current stays off the ground plane. Latch U1 (clocked by 7TRJAC, J113–13) output HIGH = string ON. The gate is fed by an NPN emitter follower from +5V through 27 R (-80 mA, positive gate current in quadrants I and IV → 4-quadrant "C" grade triac; snubberless BW/CW types cannot fire in Q IV). 10 k base pull-downs keep the strings OFF while the latch is tri-stated by BLANKING. Bits 5–7 (T5–T7) go to J111 (GPIO, CPU sheet); bit 7 also drives the optional flipper relay (DNP).

G.I. string #1 (Wht–Brn / Brn)

G.I. string #2 (Wht–Org / Org)

G.I. string #3 (Wht–Yel / Yel)

G.I. string #4 (Wht–Grn / Grn)

G.I. string #5 (Wht–Vio / Vio)

